

Training data

[illegible]

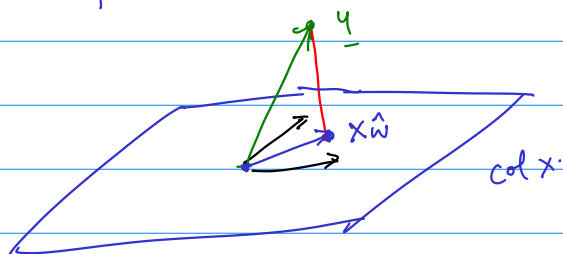
Set of all linear combinations of columns of X : $\text{col}(X)$.

$$X_{\underline{w}} : \text{If } \underline{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_4 \end{pmatrix} \quad X_{\underline{w}} = w_1 f_1 + w_2 f_2 + \dots + w_4 f_4$$

Attempt 0 Maybe y can be written as xw ?

Solve $y = x \underline{z}$. $[x \ y]$

Most likely there is no solution.



$$y - x \hat{w} = e$$

Red line orthogonal to plane: Red line is $\perp$ to every vector in plane.

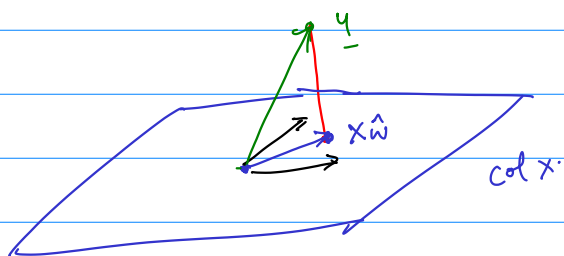
→ Suppose $M = \{v_1, \dots, v_n\}$ is a maximal linearly indep. set in $\text{col}(x)$.
(What would be a maximal lin. ind. set in $\text{col}(x)$?)

Suppose. $\underline{e}^T \underline{v}_1 = \underline{e}^T \underline{v}_2 \dots \underline{e}^T \underline{v}_n = 0.$

Pick $x \in \text{col}(x)$. Now $[v_1 \dots v_n \ x]$ has the last col free.

$$\therefore X = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$$

$$\underline{e}^T \underline{x} = \underline{e}^T \begin{bmatrix} \underline{v}_1 & \dots & \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \begin{bmatrix} \underline{e}^T \underline{v}_1 & \dots & \underline{e}^T \underline{v}_n \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = 0.$$



$$\underline{y} - \underline{\hat{x}} = \underline{e}$$

What is the closest point to $\underline{y}$? It's $\underline{\hat{x}}$ such that $\underline{y} - \underline{\hat{x}}$ is orthogonal to every pivot col of X .

Assuming all cols of X are pivot cols. $\underline{y} - \underline{\hat{x}} \perp$ every col of X .

$$X = \begin{bmatrix} | & & | \\ \underline{f}_1 & \dots & \underline{f}_{14} \\ | & & | \end{bmatrix} \quad X^T = \begin{bmatrix} - & \underline{f}_1^T & - \\ & \vdots & \\ - & \underline{f}_{14}^T & - \end{bmatrix}$$

$$X^T \underline{e} = \begin{bmatrix} - & \underline{f}_1^T & - \\ & \vdots & \\ - & \underline{f}_{14}^T & - \end{bmatrix} \underline{e} = \begin{bmatrix} \underline{f}_1^T \underline{e} \\ \vdots \\ \underline{f}_{14}^T \underline{e} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{0}$$

$X^T \underline{e} = \underline{0}$ is the formalization that $\underline{e}$ is $\perp$ to all vectors in $\text{col}(X)$.

$$\begin{aligned} \underline{0} &= X^T \underline{e} = X^T (\underline{y} - \underline{\hat{x}}) \\ &= X^T \underline{y} - X^T (\underline{\hat{x}}) \\ &= X^T \underline{y} - (X^T X) \underline{\hat{w}} \end{aligned}$$

$$X: n \times p \quad X^T: p \times n$$

$$\begin{matrix} X^T X: & p \times p \\ p \times n & n \times p \end{matrix} \quad (X^T X)^T = \underline{X^T} (X^T)^T = \underline{X^T} X$$

$$(AB)^T = B^T A^T$$

$$\begin{aligned} \left[\begin{pmatrix} | & | \\ \underline{a}_1 & \underline{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} \underline{b}_1 \\ \underline{b}_2 \end{pmatrix} \right]^T &= \left[\underline{b}_1 \underline{a}_1 + \underline{b}_2 \underline{a}_2 \right]^T = \underline{b}_1 \underline{a}_1^T + \underline{b}_2 \underline{a}_2^T \\ &= (\underline{b}_1 \quad \underline{b}_2) \begin{bmatrix} - & \underline{a}_1^T & - \\ & \underline{a}_2^T & - \end{bmatrix} \\ &= \begin{pmatrix} \underline{b}_1 \\ \underline{b}_2 \end{pmatrix}^T A^T \end{aligned}$$

$$\begin{pmatrix} a & c \\ d & b \end{pmatrix}$$

Transpose

$$\begin{pmatrix} a & d \\ c & b \end{pmatrix}$$

Therefore

$$\begin{aligned} \underline{0} &= x^T \underline{e} = x^T (y - x \hat{w}) \\ &= x^T y - x^T (x \hat{w}) \\ &= \underline{x^T y} - (x^T x) \hat{w} \end{aligned}$$

$$\underbrace{(x^T x)}_{\uparrow} \hat{w} = x^T y$$

vector of coefficients for our proposed linear model.

$x^T x$: how many pivot cols?

$x: n \times p$

$x^T x: p \times p$

Suppose $c3$ of $x = c1$ of $x + c2$ of x .

$$x^T x \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} = x^T \underline{0} = \underline{0}$$

$c4$ of $x = c1$ of $x - c2$ of x .

$$x^T x \begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \end{pmatrix} = x^T \underline{0} = \underline{0}$$

Suppose $\underline{z} \in \text{null}(x)$ $x \underline{z} = \underline{0}$

then $x^T x \underline{z} = x^T \underline{0} = \underline{0}$

$\underline{z} \in \text{null}(x^T x)$.

$$\text{null}(x) \subseteq \text{null}(x^T x)$$

_____ (1)

Suppose $x^T x \underline{v} = \underline{0}$.

then $\underline{v}^T x^T x \underline{v} = \underline{v}^T \underline{0} = 0$

$x \underline{v}$ is a vector $(x \underline{v})^T = \underline{v}^T x^T$.

$$\therefore (x \underline{v})^T x \underline{v} = 0$$

But for any vector $\underline{u}$, $\underline{u}^T \underline{u} = \|\underline{u}\|^2$.

$$\|X\underline{v}\|^2 = 0 \quad \therefore X\underline{v} = 0. \quad \therefore \text{null}(X^T X) \subseteq \text{null}(X) - (2)$$

$$\therefore \text{null}(X) = \text{null}(X^T X)$$

In particular if X has pivot in each col, $X^T X$ has too.

$$\begin{matrix} (X^T X) \\ p \times p \end{matrix} \hat{=} X^T \underline{y} \quad \hat{\underline{w}} = (X^T X)^{-1} X^T \underline{y}.$$

$$\left[\begin{array}{c} \underbrace{X^T X}_{\substack{p \text{ cols} \\ p \text{ rows}}} \quad X^T \underline{y} \end{array} \right]$$

To compute inverse of A ($n \times n$)

$$B A = I \quad \left(\text{collect all row operations to bring } A \text{ to its rref} \right)$$

We saw if $BA = I$ then $AB = I$.

$$\begin{array}{l} \text{Solve } A\underline{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \left[A \begin{array}{c} 1 \\ 0 \\ 0 \end{array} \right] \quad \left[A, I \right] \\ A\underline{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \left[A \begin{array}{c} 0 \\ 1 \\ 0 \end{array} \right] \quad \downarrow \\ \vdots \\ A\underline{x}_n = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \left[A \begin{array}{c} 0 \\ 0 \\ 1 \end{array} \right] \quad \left[I, A^{-1} \right] \end{array}$$

$$\text{If } A\underline{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad A\underline{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad A\underline{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{all have solutions.}$$

Key: A has a pivot in each of its three rows.

* A has 3 pivots.

* If A is square all solutions above are unique.

* If A is square it has an inverse.

* $A\underline{x} = \underline{b}$ has at least one soln for all $\underline{b}$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 & 1 & 1 & \underline{b} \\ 0 & 1 & 1 & 1 & \underline{b} \\ 0 & 0 & 1 & 1 & \underline{b} \end{pmatrix}$$

A has three rows

$$\left(A \mid \begin{array}{c} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{array} \right)$$

Suppose $\underline{x}$ is unique soln of $A\underline{x} = \underline{b}$ for a square A and fixed $\underline{b}$

- * $A\underline{x} = \underline{0}$ has non-zero solutions. False.
- * A has no inverse. False.
- * there is at least one $\underline{z}$ s.t. $A\underline{x} = \underline{z}$ has no soln. False. $\leftarrow$
- * the rref of $A = I$. True.
- * $A\underline{x} = \underline{c}$ has exactly one soln for all $\underline{c}$. True. $\leftarrow$

$$\begin{bmatrix} A & \underline{z} \end{bmatrix}.$$

- * If a vector $\underline{v}$ is orthogonal to every row of a matrix A .

$$A\underline{v} = \underline{0}.$$

$\Rightarrow$ another way of saying $\underline{v} \in \text{null}(A)$.

- * How to get null space of A ?

elimination on $A \rightarrow \text{rref}(A)$

$$\text{rref}(A) = \begin{array}{ccccc} & c1 & c2 & c3 & c4 & c5 \\ \begin{pmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \text{pivot} & \text{pivot} & \text{free} & \text{pivot} & \text{free}. \end{array}$$

$$c3 = -c1 + c2.$$

$$c3 + c1 - c2 = 0.$$

$$A \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \underline{0}.$$

$$c5 = 2c1 + c4.$$

$$c5 - 2c1 - c4 = 0.$$

$$A \begin{pmatrix} -2 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \underline{0}.$$

Let $A = BC$.

* # pivots in $A \leq$ # pivots in B .

* # pivots in $A \geq$ # pivots in B ?

Every col of A is a linear comb of the cols of B .

$$\text{first col of } A = B \times (\text{first col of } C)$$

$$j^{\text{th}} \text{ col of } A = B \times (j^{\text{th}} \text{ col of } C)$$

$\therefore$ first col of $A \in \text{col}(B)$

k^{th} col of $A \in \text{col}(B)$

A maximal linearly indep set in $\text{col}(B)$ = set of pivot cols of B .

Pivot cols of A : linearly indep set in $\text{col}(B)$.

We do not know if this is maximal, but we know all maximal lin indep sets in $\text{col}(B)$ have size = # pivots in B .

Def. Linearly indep: Vectors $\underline{v}_1, \dots, \underline{v}_n$ are defined to be linearly independent if $[\underline{v}_1, \dots, \underline{v}_n]$ has a pivot in each col.

Let $\{\underline{w}_1, \dots, \underline{w}_m\}$ be a linearly indep set, $\underline{w}_1, \dots, \underline{w}_m \in L$. Now $\{\underline{w}_1, \dots, \underline{w}_m\}$ is maximal linearly indep set in L if for $\underline{x} \in L$.

$\{\underline{w}_1, \dots, \underline{w}_m, \underline{x}\}$ is NOT linearly indep.

$[\underline{w}_1, \dots, \underline{w}_m, \underline{x}]$ has a free col.

Equivalent

Let $\{\underline{w}_1, \dots, \underline{w}_m\}$ be a linearly independent set in a linear space L . It is maximal if all $\underline{x} \in L$ can be written as a linear combination of $\{\underline{w}_1, \dots, \underline{w}_m\}$.

Consider $\mathbb{R}^n$ (vectors with n coordinates)

$$I_{n \times n} = \begin{pmatrix} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ 0 & & \ddots & \\ \vdots & & & 0 \\ & & & 0 & 1 \end{pmatrix}$$

n+1 cols.

$$\begin{matrix} n \text{ rows} \\ \left(\begin{array}{c|c} I_{n \times n} & \underline{x} \\ \hline & \uparrow \end{array} \right) \end{matrix}$$

has to be free.

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{col}(B).$$

Cols of B are maximal lin indpt in $\text{col}(B)$
 $\underline{11}$ are not maximal in $\mathbb{R}^3$.